

# ICT Adoption and VAT Registration among Unincorporated Enterprises in India: Analysis of Unit-Level Data

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## Abstract

Introduction of Goods and Services Tax (GST) has resulted in harmonisation of tax compliance system in India. In the new tax regime, tax compliance system has completely moved to online (digital) platforms. Given the inequality in access to digital platform or adoption of Information and Communications Technology (ICT) among marginal, small and medium enterprises in India, tax compliance may not improve in the GST regime. In a developing country like India where majority of unincorporated enterprises are in the informal sector, staying outside the formal network may restrict their economic prospects. It is worthwhile to study the relationship between access to ICT and tax (VAT) registration among unincorporated enterprises in India. Based on unit-level data of the 73rd round NSS survey, this study explores factors influencing the decision of unincorporated enterprises to take VAT/Sales Tax registration across the Indian states. The results show that adoption of ICT in terms of using internet and computers positively influence tax registration. Policy suggestions of the study may help to expand the tax base of GST by bringing informal enterprises under the tax net.

## Keywords

Unincorporated enterprises, tax registration, value added tax (VAT), information and communications technology (ICT), Probit model

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Informality in a developing country has been drawing continuous attention from academics and policymakers as it has significant implications for firm's performance, government revenues and development of the overall private sector. The extent of informality can be measured through participation of economic agents in tax regime. In India, informal sector is playing a significant role in terms of employment generation. Non-participation of informal sector in tax regime raises concerns for tax administration. Therefore, encouraging participation in the tax system by bringing more eligible taxpayers under the tax net is important to expand the tax base and also to improve tax environment (tax morale) of the country (Hofmann et al., 2008).

Participation in any tax regime starts with tax registration. However, decision of individual taxpayer to take tax registration depends on cost–benefit analysis and if benefits exceed the costs, taxpayers opt for registration. Tax registration helps the enterprises or business to become part of integrated supply chains of the formal economy (Mukherjee & Rao, 2019). Formalisation helps businesses in expanding the scale of economic operation (Demenet et al., 2016; McKenzie & Sakho, 2010) and accessing formal sources of credit (Araujo & Rodrigues, 2016; Badola & Mukherjee, 2021; de Paula & Scheinkman, 2011; Mukherjee & Rao, 2019). In spite of having several benefits associated with tax registration, a large segment of unincorporated enterprises in India do not often participate in the tax regime (Mukherjee & Rao, 2019). Unincorporated enterprises mainly include proprietary enterprises and partnership firms which together constitute 90.96% of taxpayer base in the GST regime and contribute about 20.7% of the GST collection in India (GST Council, 2020).<sup>1</sup> Given the importance of unincorporated enterprises in the GST regime, it becomes worthwhile to explore the factors influencing decision of entrepreneurs to take tax registration—particularly for value added tax (VAT). Therefore, one of the objectives of the study is to explore the VAT registration status of the unincorporated enterprises and factors associated with it.

Introduction of GST in India since 1 July 2017 has not only harmonised design, structure and administration of consumption tax system in India but also moved tax compliance system entirely to online/digital platforms to facilitate tax compliance. The GST network (GSTN) provides the common digital platform to Central and State tax administrations. Several functions of tax administration have moved to GSTN platform (e.g., tax registration, return submission, tax payment). Online tax compliance system is likely to reduce tax compliance costs. The use of digital platforms may have co-benefits to enterprises in terms of reducing information asymmetry between tax administration and taxpayers. However, not all taxpayers are equipped with adequate information and communications technology (ICT) infrastructure to comply with digital platforms (or compliance environment). Given the inequality in accessing digital compliance environment, marginal, small and medium enterprises may find it difficult to take part in the tax system, as they may have to incur higher tax compliance costs to access those services provided by third parties. If unincorporated enterprises stay outside the formal network of economic activities, it may restrict economic prospects of the country in terms of loss of economic efficiency. Therefore, this

article evaluates the adoption of ICT by unincorporated enterprises and its impact on tax compliance.

In this article, by tax compliance, we mean the registration under the State VAT/Sales Tax Act. Our analysis will also be applicable to the GST regime, as GST and VAT only differ in coverage. The article seeks to identify the factors influencing the decision of unincorporated enterprises to take VAT/Sales tax registration with a special emphasis on the role of ICT. The potential of ICT in VAT registration has never been explored in earlier literature within the Indian context. Mukherjee and Rao (2019) explore the factors associated with VAT registration among unincorporated enterprises in India. However, the study is not comprehensive in terms of coverage of factors, and especially ICT-related variables. Hence, this article presents a new dimension to the existing literature. One may also note that the digital platform or online portal for tax registration was also there in the VAT/Sales tax regime for some Indian States. Therefore, for taxpayers, digital tax compliance environment in the GST regime is not new. This article's focus will be on the number of enterprises registered, not on the amount of taxes paid by them.

In the next section, we review existing literature to identify research gaps. Following which, we discuss on data sources, data clearing process and data description. We, then, present basic statistics on factors influencing the decision of unincorporated enterprises to take VAT/sales tax registration. Then we explain the methodology, specifically heteroskedastic probit model; it is followed by discussion on results of the study and conclusions.

## Literature Review

Given the importance of tax registration under any tax regime, we explore factors influencing the decision of entrepreneurs to take tax registration based on literature review. There are two strands of literature: one explores theoretical models to understand the effects of VAT on informality; and the other explores factors associated with registration under VAT or any tax regime. The first strand of literature includes studies by Araujo and Rodrigues (2016), Boadway and Sato (2009), de Paula and Scheinkman (2010), Emran and Stiglitz (2005), Joshi et al. (2014), Kanbur and Keen (2014), Keen (2008), Keen and Mintz (2004), Piggot and Whalley (2001). The second strand of literature includes studies by Webley et al. (2002), Hansford et al. (2003), Harju et al. (2015), Klahr et al. (2017), Mukherjee and Rao (2019). Analysis of existing literature shows that there are several pull factors (e.g., productivity gain, access to formal credit, lower interest on credit) as well as push factors (e.g., annual turnover, total number of workers, size of capital asset, annual investment) which may drive the decision of entrepreneurs to register for VAT. Since our focus is to identify factors, therefore we group the variables into two categories namely push and pull factors.

First, push factors that we have identified from literature are the market value of total assets, annual turnover, and the total number of workers (Mukherjee & Rao, 2019). It has been argued that the larger the scale of operation in terms of high

turnover, workers, or assets the greater will be the chances of it becoming a formal entity. Turnover per worker was also found to be one of the most important factors (Araujo & Rodrigues, 2016; de Paula & Scheinkman, 2011; International Monetary Fund, 2017) influencing tax registration status. Also, capital productivity, labour productivity and capital-labour ratio are found to be significantly affecting the tax registration of enterprises (Mukherjee & Rao, 2019). Furthermore, the study by de Paula and Scheinkman (2011) suggests that as compared to registered enterprises, unregistered enterprises are less capital-intensive.

Second, access to formal credit (pull factor) also influences the tax registration of entrepreneurs (Araujo & Rodrigues, 2016; Badola & Mukherjee, 2021; de Paula & Scheinkman, 2011; Mukherjee & Rao, 2019). Mukherjee and Rao (2019) argue that the entrepreneurs of informal entities mostly source their credit requirements from informal creditors and for which they may have to keep transactions out of their books of accounts so that repayment of loans and interest could be facilitated in cash. This induces informal enterprises not to register under the tax regime, as with registration they have to provide their books of accounts for scrutiny before tax administration. Hence, both push and pull factors play important roles in influencing the decision of entrepreneurs of unincorporated enterprises to opt for VAT registration. There are also several enterprise-specific factors such as type of ownership, type of enterprise, year of operation, and so on, which influence the decision as well (Mukherjee & Rao, 2019). We will discuss these factors in detail in the coming sections. Moreover, the relevance and significance of factors described above within the Indian unincorporated enterprises' space are well established by Mukherjee and Rao (2019). However, the study is based on the 67th round of National Sample Survey (NSS) conducted during July 2010 to June 2011. Also, the study is not comprehensive in terms of coverage of factors, especially ICT-related variables.

The existing studies on unincorporated enterprises in India have not explored the role of access to ICT facilities on tax registration. It is expected that online registration and electronic filing (e-filing) of tax returns may facilitate ease of tax compliance and therefore encourage voluntary tax compliance, prevent corruption and tax evasion practices. Lubua (2014) examines how e-transparent services address the problem of low voluntary tax compliance by small and medium enterprises in Tanzania. The study argues that use of ICTs positively improves voluntary compliance. Access to ICT facilities such as smartphones, computers, and so on, can help in disseminating taxpayer services with respect to tax laws and tax-related activities. With the help of ICT platforms, imparting essential trainings/awareness to taxpayers on taxation system becomes easier. The study also identifies that awareness of tax is one of the factors which influence the voluntary tax compliance. Multiple studies show that lack of understanding and knowledge of tax-related issues could be plausible reasons for non-compliance (Chan et al., 2012; Cuccia, 2013; Eriksen & Fallan, 1996; Mukhlis et al., 2015; Saad, 2014). According to Saad (2014), the complexity of the tax system and lack of knowledge of tax-related issues may lead to non-compliance behaviour of the taxpayers in New Zealand. Similarly, Mukhlis et al. (2015) claim that tax education is important to enhance the tax knowledge and tax compliance in Indonesia. Dissemination of

tax-related information such as tax rates, tax payments schedule, type of taxes, and so on, is crucial to strengthening tax compliance.

Online tax registration, e-filing of tax returns, tax knowledge, and so on, could be influenced by internet penetration rate or access to internet among enterprises. One can make use of the internet with the help of computers or smartphones and could use either as a tool for registering, e-filing tax returns and to access tax-related information. Furthermore, technological improvements or access to ICT facilities may not only help in increasing taxpayers' knowledge base but also in tax compliance (OECD, 2016). With the changing technology, various tax-related services can now be accessed through online platforms. For example, a study by OECD (2016) explores the changing landscape of tax services that are traditionally provided by tax advisors, tax practitioners and tax accountants. Technological changes have now enabled a new type of services such as online tax accounting, tax filing and online tax advisory services which could be accessed from remote places via smartphones/computers with an internet connection. Providers of tax-related services are offering mostly digital or automated versions of existing services (OECD, 2016). The study by OECD (2014) describes how tax compliance could be improved with technological developments such as mobile payment services, electronic invoices, cash registers and online accounting systems. These technological developments may provide better services at a lesser cost to SMEs and could reduce their tax compliance costs. By adopting a more system-based strategy rather than a traditional case-by-case method, tax administrations could also embed taxation in corporate operations, improving efficiency in tax administration at a relatively lower tax administration costs (OECD, 2016).

However, there is another strand of literature which argues that introduction of online tax filing can increase the tax compliance costs. Tax compliance cost is often referred to as 'hidden cost of taxation' (Sandford, 1973) and acts as a disincentive for the enterprises to comply with the prevailing tax regime. e-filing of tax returns may be convenient for enterprises, but the study by Buske et al. (2018) does not show that the compliance burden reduces with e-filing in European countries. In the context of UK's VAT system, Hansford et al. (2003) find that higher compliance costs were associated with the use of the computerised method of VAT record keeping. This might be because of costs of procuring computers, staff training and software (Hansford et al., 2003). Yilmaz and Coolidge (2013) in their study on SMEs in South Africa, Ukraine and Nepal argue that in some cases e-filing does help in reducing compliance costs in South Africa; however, in Ukraine and Nepal, the tax compliance costs actually increased. Divergence between policy and practices is the major reason behind rising compliance costs in Ukraine and Nepal. Therefore, Yilmaz and Coolidge (2013) suggested that until the small businesses have regular access to computers and also become familiar with the use of internet, they should not be forced to use e-filing. But on the contrary, Rametse (2010) argues that despite incurring high costs after the implementation of GST regime, some small business in Australia are deriving the managerial benefits from using technology and better record-keeping. Small businesses have to acquire computers and software after the implementation of GST and thus they are reporting high managerial benefits (Rametse, 2020).

But the study did not explore whether the adoption of computers and software led to increase in GST registration in Australia. Furthermore, as per the study by Smulders and Evans (2017), there is mixed evidence on actual role of technology in mitigating the compliance costs of taxation in both developed and developing countries.

Therefore, in existing literature, whether access to ICT facilities leads to increase in tax compliance or tax registration is not free from ambiguity and similarly there is lack of robust evidence on the impact of technology on VAT compliance costs (Smulders & Evans, 2017). Therefore, in this article, we emphasise on the role of access to internet and computer facilities (as an indicator for ICT adoption) along with some new and established factors in determining VAT registration of unincorporated enterprises in India.

## Data Sources

In this study, we are analysing the unit level data of the 73rd round survey of NSS conducted during July 2015 to June 2016.<sup>2</sup> The total number of enterprises covered under the 73rd round survey is 290,113 of which only 18,618 enterprises (6.42%) are registered under state VAT/Sales Tax Act.

There are a few options for entrepreneurs to organise their business—registering under the Indian Companies Act is one such (known as corporate entities or companies). Other options are firms (partnership), an association of persons (AOPs), body of individuals (BOIs), proprietary enterprises, charitable entities, self-help groups, trusts and others. For the purpose of our analysis, we have restricted our sample to proprietary enterprises and partnership firms, as the tax status of the other categories (e.g., self-help groups, trusts) are not clear. Furthermore, we have restricted our analysis to the enterprises engaged only in manufacturing and trading activities. We have excluded enterprises engaged in services as majority of service providers were not required to be registered under the state VAT/sales tax regime (Mukherjee & Rao, 2019).

Since the focus of this article is on registration status, we have excluded those enterprises which are not required to be registered under the tax regime based on VAT registration threshold. In all states, enterprises with an annual turnover below a certain threshold are not legally required to be registered under VAT regime. VAT registration threshold varies across states and sectors, that is, manufacturing vis-à-vis trading (see Table A1). Since it is not mandatory for enterprises having annual turnover below the registration threshold to register under the VAT/Sales Tax Act, we have taken only those enterprises which are legally required to be registered under the State VAT/sales tax.<sup>3</sup> Moreover, we have excluded States with a few observations.

With all cleaning up of data, we have 58,896 observations of which 12,581 (21.36%) enterprises are registered and 46,315 (78.63%) are unregistered under VAT/sales tax (Table A2). This clearly shows the position of VAT/sales tax compliance among unincorporated enterprises in India during 2015–2016.

The registration status differs across States (Table A2) where the higher percentages of registration are found in Himachal Pradesh (42.37%), Kerala (39.42%), Delhi (36.40%) and Andhra Pradesh (33.31%) and the lower percentages of registration are found in Jammu & Kashmir (5.61%), Bihar (9.71%), Assam (8.18%) and West Bengal (10.02%). To capture state-specific factors influencing tax registration, we have included state dummies. Out of sample states, one state is selected to serve as base state. We have selected Andhra Pradesh as base state, based on average per capita Gross State Value Added (GSVA) at basic prices during 2014–2015 to 2016–2017. Average per capita GSVA of Andhra Pradesh is ₹110,207 which is close to the median value of ₹104,740 (Table A1).

## Factors Influencing VAT Registration

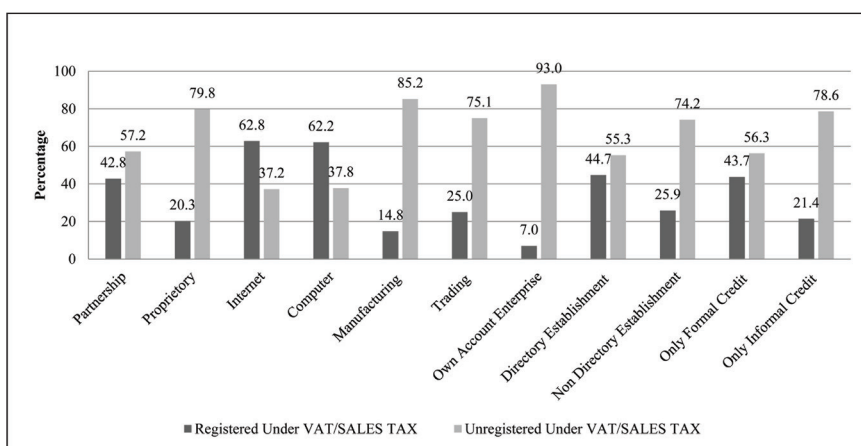
The NSS survey provides information on various economic and operational characteristics of the unincorporated non-agriculture enterprises (excluding construction). Taking a cue from the existing literature, we have identified significance of each factor associated with VAT registration of enterprises.

### Operational Characteristics

Operational characteristics include enterprise-specific factors and according to Mukherjee and Rao (2019), enterprise-specific factors like type of ownership, type of enterprise, type of economic activities, and so on, influence entrepreneurs' decision to take VAT registration. According to the NSS survey, there are mainly two types of ownership of enterprises: proprietary enterprises and partnership firms. Proprietary enterprises are run by an individual as sole owner of the business whereas partnership firms are run by more than one individual as partner and share the profits. It is expected that proprietary enterprises are often reluctant to come under the tax system as compared to partnership firms, as proprietary enterprises are considered to be too small to comply with the tax laws.

It is expected that larger the enterprise in terms of number of workers, greater will be the probability of it participating in the tax regime. Our results also suggest the same: 79.75% of proprietary enterprises are unregistered whereas 57.24% of partnership firms are unregistered under VAT (Figure 1). Similarly, 92.98% and 74.15% of the Own Account Enterprises (OAE) and Non-Directory Enterprises (NDE) are unregistered under the VAT, respectively. Of Directory Establishments (DE) 55.3% are unregistered, which shows that relatively small enterprises are more likely to stay outside the tax system.<sup>4</sup> Access to formal credit also varies across VAT registration of enterprises and registered entities are more likely to have access to formal sources of credit as compared to unregistered enterprises (Badola & Mukherjee, 2021).

We find that only 9.8% and 12.32% of sample enterprises, irrespective of whether registered under VAT/sales tax, are using internet and computer,



**Figure 1.** Broad Characteristics of the Enterprises by VAT/Sales Tax Registration Status (% of Observations)

**Source:** Computed by authors using 73rd round unit-level data.

respectively. In Delhi, Himachal Pradesh, Karnataka, Kerala and Maharashtra higher percentage of enterprises use internet and computer as compared to other states (Table A4). We find that 62% VAT registered enterprises use internet and computer as compared to 37% of VAT unregistered enterprises. To understand whether operational characteristics are significant factors influencing decision of entrepreneurs to take VAT registration, we have carried out two-sample mean equality tests.

Almost all the variables discussed above such as enterprise type, ownership type, type of economic activity (manufacturing or trading), type of credit sources found to be significant in influencing the tax registration status of the enterprises (Table 1). Now, apart from the variables already discussed by Mukherjee and Rao (2019), there are other variables such as access to ICT facilities (internet and computer),<sup>5</sup> maintenance of books of accounts and bank accounts that are found to be statistically significant. Maintenance of written statement of accounts and bank accounts reflects the transparency of an enterprise; hence, it is expected that transparent enterprises may be willing to comply with the tax regime more readily.

### *Economic Characteristics*

Size of an enterprise in terms of turnover, number of workers and market value of total assets (excluding land)<sup>6</sup> influence the decision of entrepreneur(s) to take VAT registration (Mukherjee & Rao, 2019). It is expected that larger enterprises are comparatively well-integrated with the rest of the economy and therefore their chances of participating in the tax regime is higher. Our results also indicate that registered enterprises have comparatively higher average annual turnover, average

**Table 1.** Comparison of Operational Characteristics by VAT/Sales Tax Registration Status (Average in %)

| Description                                                             | Registered | Unregistered |
|-------------------------------------------------------------------------|------------|--------------|
| Ownership type                                                          |            |              |
| Partnership firm (Yes = 1, No = 0)*                                     | 0.099      | 0.036        |
| Proprietary enterprise (Yes = 1, No = 0)*                               | 0.900      | 0.963        |
| Enterprise type                                                         |            |              |
| Own account enterprise (Yes = 1, No = 0)*                               | 0.111      | 0.401        |
| Directory establishment (Yes = 1, No = 0)*                              | 0.211      | 0.070        |
| Non-directory establishment (Yes = 1, No = 0)*                          | 0.677      | 0.527        |
| Economic activities                                                     |            |              |
| Manufacturing only (Yes = 1, No = 0)*                                   | 0.244      | 0.382        |
| Trading only (Yes = 1, No = 0)*                                         | 0.755      | 0.617        |
| ICT facilities                                                          |            |              |
| Enterprise using Internet (Yes = 1, No = 0)*                            | 0.288      | 0.046        |
| Enterprise using computer (Yes = 1, No = 0)*                            | 0.358      | 0.059        |
| Credit choice                                                           |            |              |
| Formal sources (Yes = 1, No = 0)*                                       | 0.591      | 0.384        |
| Informal sources (Yes = 1, No = 0)*                                     | 0.384      | 0.541        |
| Both (formal & informal) (Yes = 1, No = 0)*                             | 0.117      | 0.074        |
| Maintenance of account                                                  |            |              |
| Having bank account/post office savings bank account (Yes = 1, No = 0)* | 0.950      | 0.754        |
| Maintain books of account (Yes = 1, No = 0)*                            | 0.783      | 0.231        |
| Other variables                                                         |            |              |
| Assistance received from government (Yes = 1, No = 0)*                  | 0.032      | 0.011        |
| Years of operation (in year)*                                           | 12.237     | 10.893       |

**Source:** Computed by authors based on NSSO 73rd round unit-level data.

**Note:** \* implies the two-sample t-test for mean equality is significant at the 0.01 level.

market value of total assets and average number of total workers (Table 2). Average market value of total assets is 3.15 times higher for registered enterprises as compared to unregistered enterprises. Similarly, average market value of owned assets, hired assets and total workers are higher for registered enterprises as compared to unregistered enterprises. Also, the value of average annual turnover is higher for registered enterprises.

Since the scale of operation of registered enterprises is comparatively higher, it is expected that their credit requirements will also be higher. Also, registered enterprises have comparatively more access to formal credit. Therefore, it is expected that average amount of outstanding loans (credit) is higher for registered enterprises. We find that average outstanding loan of registered enterprises is 5.8 times higher than unregistered enterprises; however, there is no significant difference in average interest rates (Table 2). Table 2 also shows that average capital-labour ratio is 1.71 times higher for registered enterprises than unregistered

**Table 2.** Comparison of Economic Characteristics by VAT Registration Status

| Description                                                                    | Registered | Unregistered |
|--------------------------------------------------------------------------------|------------|--------------|
| Size of the enterprise                                                         |            |              |
| Average annual turnover (₹)*                                                   | 123,00,000 | 2,246,179    |
| Average annual gross value added (₹)*                                          | 1,334,277  | 434,131      |
| Average market value of total asset(₹)*                                        | 1,848,440  | 585,403      |
| Average market value of owned asset (₹)*                                       | 1,276,255  | 366,866      |
| Average market value of hired asset (₹)*                                       | 572,184    | 218,537      |
| Average number of total worker <sup>a</sup> (Number)*                          | 6.882      | 3.357        |
| Loan and interest rate                                                         |            |              |
| Average amount of outstanding loan (₹)*                                        | 1,528,419  | 263,011      |
| Average interest rate per month (%)                                            | 1.289      | 1.364        |
| Ratios                                                                         |            |              |
| Average gross value added per unit of turnover **                              | 0.197      | 0.290        |
| Average gross value added per unit of market value of total assets*            | 1.508      | 2.769        |
| Average gross value added per worker (₹)*                                      | 229,718    | 143,388      |
| Capital to labour ratio (average market value of total assets per worker) (₹)* | 367,808    | 214,908      |

**Source:** Computed by authors based on unit-level data of the 73rd round NSS survey.

**Notes:** # This is the ratio of annual gross value added to annual turnover.

\* implies two-sample t-test for mean equality is significant at the 0.01 level.

<sup>a</sup> The total number of workers includes full time and part-time working owner, formal hired worker, informal hired worker, helper/apprentice and other workers.

enterprises. Average gross value added per unit of turnover and average gross value added per unit of market value of total assets are higher for unregistered enterprises. This shows that value addition is higher for unregistered enterprises as compared to registered enterprises.

The above *t*-test results show factor-specific significance of operational and economic characteristics of enterprises with respect to registration status. In the next section, we estimate probit model to test concurrent significance of the identified factors as well as additional factors. In the upcoming sections, we will shed some light on how the factors influence VAT registration decision of entrepreneurs.

## Methodology

To identify factors influencing registration status of an enterprise, we have adopted heteroskedastic probit model.<sup>7</sup> Since we are working on large cross-sectional data set, possibility of having heteroskedasticity in the standard probit model cannot be ruled out. Hence, to control for the presence of heteroskedasticity in our estimated models and to derive unbiased, efficient estimates of coefficients, we have used

the heteroskedastic probit model. Heteroskedastic probit model is the generalisation of the probit model which allows the variance of the error terms in the probit model to vary as a function of independent variable(s). It is likely that the presence of continuous variables in our model specifications (e.g., market value of assets, gross value added, total number of workers) could be the reasons for the presence of heteroskedasticity. In other words, continuous variables have potential to influence the variance of the error term. We have tried with all continuous variables included in our model specifications to structure the variance of the error term and reported those in the regression models which have significant influence.

Heteroskedastic probit model (or ‘Hetprobit’) fits a maximum-likelihood probit model. The variables and the estimated model are described as follows:

$y_i$ ,  $i = 1, 2, \dots, n$ , is a binary outcome variable taking on the value 1 if success (i.e., enterprise registered under VAT/Sales Tax) and 0 if failure (i.e., enterprise unregistered under VAT/Sales Tax). In the given probit model, the probability that  $y_i$  takes on the value 1 is modelled as a nonlinear function of linear combination of the  $k$  independent variables  $x_i = (x_{i1}, x_{i2}, \dots, x_{ik})$ ,

$$Pr(y_i = 1 | x) = \Phi(x'_i \beta)$$

Where  $\beta$  represents the constant  $k$ -dimensional vector and  $\Phi()$  is a standard normal cumulative distribution function with variance 1.

By assuming  $\Phi()$  to be a cumulative distributive function with variance ( $\sigma^2$ ) which is a function of independent variables, the heteroskedastic probit model generalises the probit model. According to Harvey (1976) ‘hetprobit’ models the variance as multiplicative function of  $p$  variables  $z_i = z_{i1}, z_{i2}, z_{i3}, \dots, z_{ip}$  with  $n$  observations each.

$$\sigma_i^2 = \{\exp(z_i^2 \gamma)\}^2$$

Hence, the success of probability as a function of all independent variables can be written as follows:

$$Pr(y_i = 1 | x, z) = \Phi\left(\frac{x'_i \beta}{\exp(z'_i \gamma)}\right) \quad (1)$$

Furthermore, let  $y$  be the  $n \times 1$  vector of all observations of  $y_i$ , let  $X$  be the  $n \times k$  matrix whose  $i^{\text{th}}$  row is  $x'_i$  and finally  $Z$  to be the  $n \times p$  matrix with  $i^{\text{th}}$  row of  $z'_i$ . Therefore, now the log-likelihood function can be written in the following form.

$$\ln L(\beta, \gamma, |y, X, Z) = \sum_{i=1}^n y_i \ln \Phi\left(\frac{x'_i \beta}{\exp(z'_i \gamma)}\right) + (1 - y_i) \ln \left(1 - \Phi\left(\frac{x'_i \beta}{\exp(z'_i \gamma)}\right)\right).$$

To get maximum-likelihood estimate given this set of  $n$  observed values of  $y$ ,  $x$ ,  $z$  random variables, one has to maximise this function over the possible choices of  $(\beta, \gamma) \in R^{k+p}$ .

We have run the following binary choice heteroskedastic probit models by using the following specifications:

$$Pr(vatact = 1 | x, z) = \Phi\left(\frac{x'_i\beta}{\exp(z'_i\gamma)}\right)$$

### Model 1 specification

$$\begin{aligned} x'\beta &= \beta_0 + \beta_1 \text{Inturn} + \beta_2 \text{Intotwork} + \beta_3 \text{lnmktasset} + \beta_4 \text{yooop} + \beta_5 \text{locationout*urban} + \\ &\beta_6 \text{manuf} + \beta_7 \text{prop*oe} + \beta_8 \text{mainacc} + \beta_9 \text{bankacc} + \beta_{10} \text{internet} + \beta_{11} \text{computer} + \\ &\beta_{12} \text{state dummies} \\ z'\gamma &= \gamma_0 \text{Inturn} + \gamma_1 \text{lnmktasset} \end{aligned}$$

### Model 2 specification

$$\begin{aligned} x'\beta &= \beta_0 + \beta_1 \text{Inturn} + \beta_2 \text{lnmktasset} + \beta_3 \text{yooop} + \beta_4 \text{locationout*urban} + \beta_5 \text{manuf} + \\ &\beta_6 \text{prop*oe} + \beta_7 \text{franchise} + \beta_8 \text{mainacc} + \beta_9 \text{bankacc} + \beta_{10} \text{internet} + \beta_{11} \text{computer} + \\ &\beta_{12} \text{gvacapital} + \beta_{13} \text{state dummies} \\ z'\gamma &= \gamma_0 \text{Inturn} + \gamma_1 \text{lnmktasset} \end{aligned}$$

### Model 3 specification

$$\begin{aligned} x'\beta &= \beta_0 + \beta_1 \text{Intotwork} + \beta_2 \text{yooop} + \beta_3 \text{locationout*urban} + \beta_4 \text{manuf} + \beta_5 \text{prop*oe} + \\ &\beta_6 \text{franchise} + \beta_7 \text{formalcred} + \beta_8 \text{mainacc} + \beta_9 \text{bankacc} + \beta_{10} \text{internet} + \beta_{11} \text{computer} + \\ &\beta_{12} \text{gvacapital} + \beta_{13} \text{state dummies} \\ z'\gamma &= \gamma_0 \text{Intotwork} \end{aligned}$$

### Dependent variable

*vatact*: 1 if the enterprise registered under VAT/Sales Tax, 0 otherwise.

### Economic characteristics (scale factors)

- Inturn*: log of annual turnover (in ₹)  
*Intotwork*: log of total workers (full-time and part-time, male, female and transgender)  
*lnmktasset*: log of the market value of total (owned and hired) assets excluding land (in ₹)

### Operational characteristics (enterprise-specific factors)

- yooop*: years of operation (as on 2016)  
*locationout\*urban*: 1 if the location of the enterprise is outside household premises and in urban area, 0 otherwise.  
*manuf*: 1 if the enterprise is engaged only in manufacturing activities, 0 otherwise (trading activity form the reference category for activity type).  
*prop\*oe*: 1 if the enterprise is proprietary and own account enterprise, 0 otherwise.  
*franchise*: 1 if the enterprise is franchisee outlet, 0 otherwise.

*formalcred*: 1 if the enterprise has taken credit only from formal sources,<sup>8</sup> 0 otherwise.

*mainacc*: 1 if the enterprise maintains books of accounts, 0 otherwise.

*bankacc*: 1 if the enterprise maintains any bank/post office savings bank account, 0 otherwise

*internet*: 1 if the enterprise uses the internet during the last 365 days and 0 otherwise.

*computer*: 1 if the enterprise uses the computer during the last 365 days and 0 otherwise.

*gvacapital*: ratio of annual gross value added and market value of total assets.

### State-specific factor

*State Dummy*: 1 for the relevant state, 0 otherwise

The regression results and basic statistics are given in Table 3 and Table A5, respectively. Basic statistics of underlying variables are presented in Table A5.

## Results and Discussions

Scale factors are positively and significantly influencing the VAT registration status. **Own-account (proprietary) enterprises as well as manufacturing enterprises are less likely to register under the tax regime.** The findings are as per our expectations: when the scale of economic activities of an enterprise increases, so does the chance of integrating with the larger supply chains of the economy via tax registration. Manufacturing enterprises are less likely to register under VAT vis-à-vis enterprises engaged in trading. Perhaps, manufacturing enterprises are not well-integrated with the larger supply chains; they are small in size and have either their own marketing system (e.g., direct sales to consumers) or sales to small traders who are also unregistered. According to Mukherjee and Rao (2019), the input tax credit claims of manufacturing enterprises are small as a result the self-policing property of VAT is relatively weak for manufacturers. Despite having a lower VAT threshold as compared to enterprises engaged in trading, the manufacturing enterprises are relatively less likely to register with the tax system.

Enterprises that have access to credit from formal sources are likely to be registered under the tax regime. Alternatively, VAT registration helps enterprises to secure formal credit. Enterprises having more gross value added (GVA) per unit of market value of total asset are likely to be unregistered. These results are in line with the existing study by Mukherjee and Rao (2019).

Enterprises operating for a longer time are more likely to be registered under VAT/Sales Tax Act. This indicates that new enterprises or businesses are less likely to come under the tax regime. Similarly, enterprises which are more transparent in terms of maintaining books of accounts and bank accounts are more likely to be registered under VAT/Sales Tax Act. One more interesting enterprise-specific factor turns out to be positive and significant is franchise.

Table 3. Regression Results

| Variable Name              | Model 1 |              |        | Model 2 |              |        | Model 3 |              |        |
|----------------------------|---------|--------------|--------|---------|--------------|--------|---------|--------------|--------|
|                            | Coef.   | S.E.         | M.E.   | Coef.   | S.E.         | M.E.   | Coef.   | S.E.         | M.E.   |
| <i>Inturn</i>              | 1.187   | ***<br>0.142 | 0.084  | 1.205   | ***<br>0.144 | 0.086  |         |              |        |
| <i>Intotwork</i>           | 0.096   | *<br>0.054   | 0.006  |         |              |        | 0.287   | ***<br>0.029 | 0.099  |
| <i>lnmktasset</i>          | 0.381   | ***<br>0.049 | 0.031  | 0.372   | ***<br>0.050 | 0.029  |         |              |        |
| <i>yoop</i>                | 0.011   | ***<br>0.003 | 0.001  | 0.011   | ***<br>0.003 | 0.001  | 0.008   | ***<br>0.002 | 0.002  |
| <i>locationout*urban</i>   | 0.544   | ***<br>0.083 | 0.032  | 0.531   | ***<br>0.081 | 0.032  | 0.222   | ***<br>0.035 | 0.067  |
| <i>manuf</i>               | -1.352  | ***<br>0.173 | -0.075 | -1.259  | ***<br>0.159 | -0.070 | -0.686  | ***<br>0.048 | -0.190 |
| <i>prop*oe</i>             | -0.841  | ***<br>0.115 | -0.047 | -0.898  | ***<br>0.118 | -0.051 | -0.249  | ***<br>0.053 | -0.071 |
| <i>franchise</i>           |         |              |        | 0.613   | ***<br>0.200 | 0.040  | 0.260   | **<br>0.107  | 0.083  |
| <i>formalcred</i>          |         |              |        |         |              |        | 0.140   | ***<br>0.037 | 0.042  |
| <i>mainacc</i>             | 3.098   | ***<br>0.358 | 0.212  | 3.080   | ***<br>0.357 | 0.211  | 1.102   | ***<br>0.048 | 0.329  |
| <i>bankacc</i>             | 1.168   | ***<br>0.153 | 0.061  | 1.163   | ***<br>0.153 | 0.061  | 0.538   | ***<br>0.071 | 0.139  |
| <i>internet</i>            | 0.376   | ***<br>0.132 | 0.023  | 0.362   | **<br>0.130  | 0.023  | 0.344   | ***<br>0.070 | 0.110  |
| <i>computer</i>            | 1.332   | ***<br>0.198 | 0.092  | 1.314   | ***<br>0.196 | 0.091  | 0.320   | ***<br>0.064 | 0.101  |
| <i>gvacapital</i>          |         |              |        | -0.013  | **<br>0.007  | -0.001 | -0.022  | ***<br>0.005 | -0.006 |
| <i>Jammu &amp; Kashmir</i> | -1.950  | ***<br>0.288 | -0.086 | -1.962  | ***<br>0.289 | -0.086 | -0.716  | ***<br>0.118 | -0.166 |
| <i>Himachal Pradesh</i>    | 1.104   | ***<br>0.244 | 0.077  | 1.054   | ***<br>0.239 | 0.073  | 0.165   | 0.101        | 0.052  |
| <i>Punjab</i>              | -1.641  | ***<br>0.253 | -0.075 | -1.663  | ***<br>0.255 | -0.076 | -0.329  | ***<br>0.102 | -0.089 |
| <i>Uttarakhand</i>         | -0.692  | ***<br>0.225 | -0.036 | -0.710  | ***<br>0.224 | -0.037 | -0.069  | 0.134        | -0.020 |
| <i>Haryana</i>             | -0.970  | ***<br>0.202 | -0.049 | -0.989  | ***<br>0.203 | -0.050 | -0.054  | 0.099        | -0.016 |
| <i>Delhi</i>               | -1.286  | ***<br>0.228 | -0.062 | -1.281  | ***<br>0.228 | -0.062 | -0.200  | *<br>0.118   | -0.056 |
| <i>Rajasthan</i>           | -1.253  | ***<br>0.228 | -0.061 | -1.278  | ***<br>0.229 | -0.062 | -0.062  | 0.106        | -0.018 |
| <i>Uttar Pradesh</i>       | -0.095  | 0.153        | -0.006 | -0.111  | 0.153        | -0.006 | 0.071   | 0.086        | 0.021  |

|                                                 |           |     |        |        |           |       |        |        |           |     |        |        |
|-------------------------------------------------|-----------|-----|--------|--------|-----------|-------|--------|--------|-----------|-----|--------|--------|
| Bihar                                           | -1.912    | *** | 0.288  | -0.083 | -1.939    | ***   | 0.291  | -0.084 | -0.460    | *** | 0.143  | -0.118 |
| Assam                                           | -2.584    | *** | 0.357  | -0.099 | -2.598    | ***   | 0.359  | -0.099 | -0.781    | *** | 0.165  | -0.174 |
| West Bengal                                     | -2.890    | *** | 0.368  | -0.113 | -2.879    | ***   | 0.368  | -0.113 | -0.840    | *** | 0.087  | -0.198 |
| Orissa                                          | -0.991    | *** | 0.242  | -0.050 | -1.013    | ***   | 0.242  | -0.051 | -0.294    | *** | 0.112  | -0.080 |
| Chhattisgarh                                    | -2.389    | *** | 0.350  | -0.095 | -2.394    | ***   | 0.351  | -0.095 | -0.775    | *** | 0.140  | -0.174 |
| Madhya Pradesh                                  | -1.610    | *** | 0.246  | -0.074 | -1.619    | ***   | 0.247  | -0.074 | -0.123    | *** | 0.129  | -0.035 |
| Gujarat                                         | -2.061    | *** | 0.293  | -0.088 | -2.049    | ***   | 0.292  | -0.088 | -0.667    | *** | 0.102  | -0.159 |
| Maharashtra                                     | -2.793    | *** | 0.361  | -0.107 | -2.788    | ***   | 0.362  | -0.107 | -0.573    | *** | 0.084  | -0.144 |
| Karnataka                                       | -0.945    | *** | 0.204  | -0.048 | -0.949    | ***   | 0.204  | -0.048 | -0.261    | *** | 0.083  | -0.072 |
| Kerala                                          | -0.479    | *** | 0.184  | -0.026 | -0.507    | *     | 0.184  | -0.028 | -0.060    | *** | 0.081  | -0.018 |
| Tamil Nadu                                      | -1.426    | *** | 0.234  | -0.068 | -1.422    | ***   | 0.234  | -0.068 | -0.445    | *** | 0.079  | -0.118 |
| _cons                                           | -26.257   | *** | 2.983  |        | -26.249   | ***   | 3.006  |        | -1.714    | *** | 0.103  |        |
| Insigma <sup>2</sup>                            |           |     |        |        |           |       |        |        |           |     |        |        |
| Inturn                                          | 0.057     | *** | 0.008  |        | 0.060     | 0.008 | ***    |        |           |     |        |        |
| lnmktasset                                      | 0.034     | *** | 0.008  |        | 0.030     | 0.008 | ***    |        |           |     |        |        |
| Intotwork                                       |           |     |        |        |           |       |        |        | 0.060     | *** | 0.025  |        |
| Number of observations                          | 58,765    |     |        |        | 58,765    |       |        |        | 10,606    |     |        |        |
| Wald chi <sup>2</sup>                           | 90.97     | *** | df: 30 |        | 91.03     | ***   | df: 31 |        | 1,053.100 | *** | df: 31 |        |
| Log likelihood                                  | -19,436.5 |     |        |        | -19,430.6 |       |        |        | -4512.204 |     |        |        |
| Pseudo R <sup>2</sup>                           | 0.12473   |     |        |        | 0.124324  |       |        |        | 0.256     |     |        |        |
| LR chi <sup>2</sup> for Ho:lnsigma <sup>2</sup> | 134.76    | *** | df: 2  |        | 133.91    | ***   | df: 2  |        | 6.100     | **  | df: 1  |        |

**Source:** Computed by authors based on NSSO 73rd round unit-level data.

**Note:** \*\*\*, \*\*, \* imply the significance level of the estimated z statistic at the 0.01, 0.05 and 0.10 levels, respectively. SE is standard error; ME is marginal effect; df is degree of freedom.

The enterprises which are franchisee of a business are more likely to be registered under the tax regime.

Moving towards ICT-specific variables, our analysis shows that enterprises which are using internet and computer are more likely to be registered under the tax regime. This shows that facility to access ICT-platforms may bring enterprises under the tax net. It is likely that access to internet could reduce information asymmetries for both taxpayers and tax administration which may increase tax compliance. Similarly, access to computer can act as a tool to make use of internet for tax registration, e-filing tax returns and reducing information asymmetries. We have also run various alternative specifications of the regression model (Table A6) and our results hold across model specifications. GVA per turnover was found to be negative and significant, and it implies that the higher the value addition per unit of turnover, the greater will be the tax burden on output vis-à-vis available input tax credit, hence less incentive for entrepreneurs to take tax registration (Table A6).

As regards state-specific dummies where we have considered Andhra Pradesh as a base state, the results from all three models show that enterprises located in states such as Jammu and Kashmir, Uttarakhand, Delhi, Bihar, Assam, West Bengal, Orissa, Chhattisgarh, Gujarat, Maharashtra, Karnataka, Tamil Nadu are less likely to be registered under the VAT tax regime compared to the enterprises located in Andhra Pradesh. On the other hand, enterprises located in Himachal Pradesh are more likely to be registered under VAT/Sales tax regime as compared to the enterprises located in Andhra Pradesh.

## **Conclusions**

The study shows that only 21.36% eligible unincorporated enterprises are registered under the VAT/Sales Tax Act in 2015–2016. There is inter-state disparity in VAT registration compliance. Higher percentage of registration is observed for Himachal Pradesh (42.37%), Kerala (39.42%), Delhi (36.40%) and Andhra Pradesh (33.31%) and lower percentage of registration is observed for Jammu & Kashmir (5.61%), Bihar (9.71%), Assam (8.18%) and West Bengal (10.02%).

The study explores the role of access to internet and computer facilities of unincorporated enterprises in VAT registration along with other drivers (push and pull factors). Our analysis shows that enterprises which are using internet and computer are more likely to opt for VAT registration. This shows that access to ICT-platform increases tax compliance in terms of taking part in the tax system. However, it is disappointing to note that only 9.8% and 12.32% of sample enterprises were using internet and computer facilities respectively. With the introduction of GST, the entire tax compliance system has moved to online or digital platform. Therefore, low penetration rate of internet and access to computer facilities are likely to increase compliance costs of small taxpayers. There exists an inter-state disparity in access to internet and computer facilities. State-specific analysis shows that enterprises located in Assam, Bihar, Punjab, Uttar Pradesh and West Bengal have

less access to internet and computer. Therefore, these states may consider special policies to facilitate access to internet and computer to small taxpayers to increase tax compliance. Moreover, we have also found that there are several other factors influencing decision of entrepreneurs to take VAT registration. Factors specific to 'scale of operation' such as total workers, annual turnover and market value of total assets positively and significantly influence VAT registration. Also access to formal sources of credit persuades enterprises to register under VAT. Therefore, facilitating access to formal credit by the government may benefit enterprises to scale up their economic activities and also to integrate with larger supply chains of the economy. Furthermore, we have found that enterprises maintaining books of accounts and having bank accounts are more likely to register under VAT. Hence, transparency in business accounting is one of the factors influencing VAT registration. However, majority of unincorporated enterprises prefer not to reveal their economic activity before tax administration by not maintaining written books of accounts. The results show that the enterprises operating for a longer time are more likely to be registered under VAT/Sales Tax Act. Similarly, enterprises which are franchisee of a business are more likely to be registered under the tax regime.

We have also found that manufacturing enterprises are less likely to register under VAT as compared to enterprises engaged in trading. One possible reason could be manufacturing enterprises are not well integrated with the larger supply chains and have their own marketing system and/or supplying their outputs to unregistered traders/manufacturers. Perhaps the self-policing aspect of state VAT system, as incentivised by allowing input-tax credit against purchases of inputs and capital goods, does not encourage entrepreneurs engaged in manufacturing to opt for VAT registration. This similar result was also found in the earlier analysis by Mukherjee and Rao (2019). So, situation has not yet improved, and steps should be taken to integrate the manufacturing enterprises with the rest of the economy. Efficient tax administration by conducting regular surveys to identify errant taxpayers could help in bringing more enterprises under the tax regime.

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## Appendix

**Table A1.** VAT Threshold Values Across the Selected States of India

| State Name          | VAT Registration Threshold<br>(₹0.1 Million) |        | Average per Capita GSVA<br>at Basic Price (₹): From<br>2014–2015 to 2016–2017<br>(Current Prices) |
|---------------------|----------------------------------------------|--------|---------------------------------------------------------------------------------------------------|
|                     | Manufacturer                                 | Trader |                                                                                                   |
| Andhra Pradesh      | 7.5                                          | 7.5    | 110,207                                                                                           |
| Assam               | 6                                            | 6      | 64,061                                                                                            |
| Bihar               | 5                                            | 5      | 32,978                                                                                            |
| Chhattisgarh        | 2                                            | 10     | 80,178                                                                                            |
| Delhi               | 10                                           | 10     | 259,651                                                                                           |
| Gujarat             | 2                                            | 5      | 144,135                                                                                           |
| Haryana             | 5                                            | 5      | 163,916                                                                                           |
| Himachal Pradesh    | 4                                            | 6      | 150,143                                                                                           |
| Jammu & Kashmir     | 0                                            | 0      | 79,751                                                                                            |
| Karnataka           | 7.5                                          | 7.5    | 148,298                                                                                           |
| Kerala              | 10                                           | 10     | 151,352                                                                                           |
| Madhya Pradesh      | 5                                            | 5      | 66,857                                                                                            |
| Maharashtra         | 10                                           | 10     | 147,047                                                                                           |
| Odisha              | 10                                           | 10     | 73,939                                                                                            |
| Punjab              | 1                                            | 5      | 120,827                                                                                           |
| Rajasthan           | 5                                            | 10     | 88,652                                                                                            |
| Tamil Nadu          | 10                                           | 10     | 145,247                                                                                           |
| Uttar Pradesh       | 5                                            | 5      | 49,817                                                                                            |
| Uttarakhand         | 5                                            | 5      | 153,407                                                                                           |
| West Bengal         | 5                                            | 5      | 78,400                                                                                            |
| Minimum             |                                              |        | 32,978                                                                                            |
| Quartile 1          |                                              |        | 72,168                                                                                            |
| Quartile 2 (Median) |                                              |        | 104,740                                                                                           |
| Quartile 3          |                                              |        | 148,759                                                                                           |
| Maximum             |                                              |        | 259,651                                                                                           |

**Source:** Threshold values are taken from Mukherjee and Rao (2019) and figures for Gross State Value Added (GSVA) are taken from Ministry of Statistics and Programme Implementation.

**Table A2.** Enterprises with Annual Turnover Above the VAT Registration Threshold

| State Name     | Enterprises<br>Registered<br>Under VAT | % of<br>Enterprises<br>Registered<br>Under VAT | Enterprises<br>Not<br>Registered<br>Under VAT | % of<br>Enterprises<br>Not<br>Registered<br>Under VAT | Total |
|----------------|----------------------------------------|------------------------------------------------|-----------------------------------------------|-------------------------------------------------------|-------|
|                |                                        |                                                |                                               |                                                       |       |
| Andhra Pradesh | 601                                    | 33.31                                          | 1,203                                         | 66.69                                                 | 1,804 |
| Assam          | 155                                    | 8.18                                           | 1,741                                         | 91.82                                                 | 1,896 |
| Bihar          | 275                                    | 9.71                                           | 2,557                                         | 90.29                                                 | 2,832 |

(Table A2 continued)

(Table A2 continued)

| State Name       | Enterprises<br>Registered<br>Under VAT | % of<br>Enterprises<br>Registered<br>Under VAT | Enterprises<br>Not<br>Registered<br>Under VAT | % of<br>Enterprises<br>Not<br>Registered<br>Under VAT | Total  |
|------------------|----------------------------------------|------------------------------------------------|-----------------------------------------------|-------------------------------------------------------|--------|
| Chhattisgarh     | 163                                    | 7.70                                           | 1,954                                         | 92.30                                                 | 2,117  |
| Delhi            | 947                                    | 36.40                                          | 1,655                                         | 63.60                                                 | 2,602  |
| Gujarat          | 639                                    | 18.39                                          | 2,835                                         | 81.61                                                 | 3,474  |
| Haryana          | 739                                    | 24.20                                          | 2,315                                         | 75.80                                                 | 3,054  |
| Himachal Pradesh | 522                                    | 42.37                                          | 710                                           | 57.63                                                 | 1,232  |
| Jammu & Kashmir  | 238                                    | 5.61                                           | 4,008                                         | 94.39                                                 | 4,246  |
| Karnataka        | 850                                    | 31.69                                          | 1,832                                         | 68.31                                                 | 2,682  |
| Kerala           | 963                                    | 39.42                                          | 1,480                                         | 60.58                                                 | 2,443  |
| Madhya Pradesh   | 605                                    | 19.27                                          | 2,534                                         | 80.73                                                 | 3,139  |
| Maharashtra      | 1,081                                  | 27.69                                          | 2,823                                         | 72.31                                                 | 3,904  |
| Odisha           | 273                                    | 24.44                                          | 844                                           | 75.56                                                 | 1,117  |
| Punjab           | 569                                    | 15.35                                          | 3,137                                         | 84.65                                                 | 3,706  |
| Rajasthan        | 761                                    | 30.31                                          | 1,750                                         | 69.69                                                 | 2,511  |
| Tamil Nadu       | 965                                    | 28.19                                          | 2,458                                         | 71.81                                                 | 3,423  |
| Uttar Pradesh    | 1,387                                  | 23.70                                          | 4,465                                         | 76.30                                                 | 5,852  |
| Uttarakhand      | 290                                    | 22.43                                          | 1,003                                         | 77.57                                                 | 1,293  |
| West Bengal      | 558                                    | 10.02                                          | 5,011                                         | 89.98                                                 | 5,569  |
| Total            | 12,581                                 | 21.36                                          | 46,315                                        | 78.64                                                 | 58,896 |

**Source:** Calculated by authors using NSSO 73rd round unit-level data (2015–2016).

**Table A3.** Enterprises with Annual Turnover Below the VAT Registration Threshold

| State Name       | Enterprises<br>Registered<br>Under VAT | % of<br>Enterprises<br>Registered<br>Under VAT | Enterprises<br>not<br>Registered<br>Under VAT | % of<br>Enterprises<br>not<br>Registered<br>Under VAT | Total |
|------------------|----------------------------------------|------------------------------------------------|-----------------------------------------------|-------------------------------------------------------|-------|
| Andhra Pradesh   | 75                                     | 1.34                                           | 5,507                                         | 98.66                                                 | 5,582 |
| Assam            | 12                                     | 0.37                                           | 3,189                                         | 99.63                                                 | 3,201 |
| Bihar            | 3                                      | 0.07                                           | 4,354                                         | 99.93                                                 | 4,357 |
| Chhattisgarh     | 1                                      | 0.09                                           | 1,107                                         | 99.91                                                 | 1,108 |
| Delhi            | 49                                     | 2.78                                           | 1,715                                         | 97.22                                                 | 1,764 |
| Gujarat          | 13                                     | 0.39                                           | 3,313                                         | 99.61                                                 | 3,326 |
| Haryana          | 71                                     | 2.88                                           | 2,398                                         | 97.12                                                 | 2,469 |
| Himachal Pradesh | 101                                    | 6.86                                           | 1,372                                         | 93.14                                                 | 1,473 |
| Jammu & Kashmir  | 0                                      | 0                                              | 0                                             | 0                                                     | 0     |
| Karnataka        | 68                                     | 1.38                                           | 4,873                                         | 98.62                                                 | 4,941 |
| Kerala           | 170                                    | 3.67                                           | 4,457                                         | 96.33                                                 | 4,627 |
| Madhya Pradesh   | 33                                     | 0.46                                           | 7,137                                         | 99.54                                                 | 7,170 |

(Table A3 continued)

(Table A3 continued)

| State Name    | Enterprises Registered Under VAT | % of Enterprises Registered Under VAT | Enterprises not Registered Under VAT | % of Enterprises not Registered Under VAT | Total  |
|---------------|----------------------------------|---------------------------------------|--------------------------------------|-------------------------------------------|--------|
| Maharashtra   | 128                              | 1.69                                  | 7,452                                | 98.31                                     | 7,580  |
| Odisha        | 32                               | 0.71                                  | 4,446                                | 99.29                                     | 4,478  |
| Punjab        | 19                               | 1.22                                  | 1,535                                | 98.78                                     | 1,554  |
| Rajasthan     | 104                              | 2.24                                  | 4,543                                | 97.76                                     | 4,647  |
| Tamil Nadu    | 156                              | 1.66                                  | 9,230                                | 98.34                                     | 9,386  |
| Uttar Pradesh | 123                              | 1.11                                  | 11,007                               | 98.89                                     | 11,130 |
| Uttarakhand   | 21                               | 1.29                                  | 1,604                                | 98.71                                     | 1,625  |
| West Bengal   | 40                               | 0.57                                  | 6,927                                | 99.43                                     | 6,967  |
| Total         | 1,219                            | 1.39                                  | 86,166                               | 98.61                                     | 87,385 |

**Source:** Calculated by authors using NSSO 73rd round unit level data (2015–2016).

**Table A4.** Distribution of Enterprises Having Access to Internet and Computer Across Indian States

| State Name       | % of Enterprises Having Access to Internet <sup>#</sup> | % of Enterprises Having Access to Computer |
|------------------|---------------------------------------------------------|--------------------------------------------|
| Andhra Pradesh   | 9.42                                                    | 12.47                                      |
| Assam            | 5.49                                                    | 7.54                                       |
| Bihar            | 4.77                                                    | 5.58                                       |
| Chhattisgarh     | 4.44                                                    | 6.38                                       |
| Delhi            | 20.56                                                   | 20.75                                      |
| Gujarat          | 9.21                                                    | 12.69                                      |
| Haryana          | 10.67                                                   | 10.87                                      |
| Himachal Pradesh | 11.53                                                   | 15.67                                      |
| Jammu & Kashmir  | 2.85                                                    | 3.39                                       |
| Karnataka        | 15.29                                                   | 18.38                                      |
| Kerala           | 20.63                                                   | 29.23                                      |
| Madhya Pradesh   | 9.17                                                    | 11.88                                      |
| Maharashtra      | 17.57                                                   | 23.03                                      |
| Odisha           | 9.31                                                    | 12.09                                      |
| Punjab           | 6.88                                                    | 8.15                                       |
| Rajasthan        | 11.99                                                   | 14.85                                      |
| Tamil Nadu       | 14.23                                                   | 20.16                                      |
| Uttar Pradesh    | 5.78                                                    | 7.66                                       |
| Uttarakhand      | 13.61                                                   | 12.06                                      |
| West Bengal      | 4.99                                                    | 6.46                                       |
| Total            | 9.8                                                     | 12.32                                      |

**Source:** Computed by authors using NSSO 73rd round unit-level data (2015–2016).

**Note:** # Having access to internet or computer means the use of internet or computer by the enterprises during last 365 days.

**Table A5.** Basic Statistics

| Variable Name                                                                                                             | Mean  | Standard<br>Deviation | Min    | Max       |
|---------------------------------------------------------------------------------------------------------------------------|-------|-----------------------|--------|-----------|
| Whether the enterprise is registered under VAT/Sales Tax Act? (vatact, yes = 1; 0 = no)                                   | 0.21  | 0.41                  | 0.00   | 1.00      |
| Years of operation as of 2015 since the year of initial operation (yooop) (years)                                         | 11.18 | 8.94                  | 0.00   | 128.00    |
| Whether the enterprise located outside the household premises but within urban area? (locationout*urban, yes = 1; no = 2) | 0.48  | 0.50                  | 0.00   | 1.00      |
| Whether the enterprise is engaged in manufacturing activity? (manuf, yes = 1; no = 0)                                     | 0.35  | 0.48                  | 0.00   | 1.00      |
| Whether the enterprise is proprietary and own account enterprises during last 365 days? (prop*oae)                        | 0.33  | 0.47                  | 0.00   | 1.00      |
| Whether the enterprise is franchise outlet? (franchise, yes = 1; no = 0)                                                  | 0.02  | 0.13                  | 0.00   | 1.00      |
| Whether the enterprise maintain any bank account/post office savings bank account? (bankacc; yes = 1, no = 0)             | 0.80  | 0.40                  | 0.00   | 1.00      |
| Whether written accounts maintain by enterprises? (mainacc, yes = 1; no = 0)                                              | 0.35  | 0.48                  | 0.00   | 1.00      |
| Whether the enterprise use internet during the last 365 days? (internet; yes = 1; no = 0)                                 | 0.10  | 0.30                  | 0.00   | 1.00      |
| Whether the enterprise use computer during the last 365 days? (computer; yes = 1; no = 0)                                 | 0.12  | 0.33                  | 0.00   | 1.00      |
| Whether the enterprise has outstanding loan from formal credit sources? (formalcred; yes = 1, no = 0)                     | 0.50  | 0.50                  | 0.00   | 1.00      |
| Log of annual turnover (Inturn in ₹)                                                                                      | 14.29 | 1.18                  | 8.01   | 20.83     |
| Average annual turnover per market value of total asset, i.e., output per capital (outputcapital)                         | 15.19 | 94.91                 | 0.01   | 10,752.00 |
| Average annual gross value added per market value of total asset (gvacapital)                                             | 2.50  | 15.49                 | -18.21 | 2,305.60  |
| Average annual gross value added per annual turnover (gvturn)                                                             | 0.27  | 0.23                  | -19.18 | 1.00      |
| Log of total number of workers (Intotwork)                                                                                | 0.97  | 0.77                  | 0.00   | 6.32      |
| Log of market value of total assets (lnmktasset)                                                                          | 12.70 | 1.36                  | 4.61   | 19.01     |

**Source:** Calculated by authors using NSSO 73rd round unit-level data

**Table A6.** Alternative Regression Models

| VAT                              | Model/A1 |       |        | Model/A2 |       |        | Model/A3 |       |        |
|----------------------------------|----------|-------|--------|----------|-------|--------|----------|-------|--------|
|                                  | Coef.    | S.E.  | M.E.   | Coef.    | S.E.  | M.E.   | Coef.    | S.E.  | M.E.   |
| <i>Inturn</i>                    |          |       |        |          |       |        |          |       |        |
| <i>Intotwork</i>                 | 0.518    | 0.058 | 0.054  | 1.012    | 0.112 | 0.094  |          |       |        |
| <i>lnmktasset</i>                | 0.290    | 0.025 | 0.043  | 0.138    | 0.047 | 0.018  | 0.287    | 0.029 | 0.099  |
| <i>yoop</i>                      | 0.008    | 0.002 | 0.001  | 0.008    | 0.002 | 0.001  | 0.008    | 0.002 | 0.002  |
| <i>locationout*urban</i>         | 0.370    | 0.046 | 0.039  | 0.498    | 0.068 | 0.039  | 0.222    | 0.035 | 0.067  |
| <i>manuf</i>                     | -0.759   | 0.081 | -0.075 | -0.923   | 0.115 | -0.068 | -0.686   | 0.048 | -0.190 |
| <i>prop*oe</i>                   | -0.523   | 0.059 | -0.052 | -0.620   | 0.093 | -0.046 | -0.249   | 0.053 | -0.071 |
| <i>franchise</i>                 |          |       |        | 0.490    | 0.151 | 0.042  | 0.260    | 0.107 | 0.083  |
| <i>govtassis<sup>a</sup></i>     |          |       |        | 0.285    | 0.142 | 0.024  |          |       |        |
| <i>mainacc</i>                   | 1.808    | 0.168 | 0.222  | 2.370    | 0.266 | 0.215  | 1.102    | 0.048 | 0.329  |
| <i>bankacc</i>                   | 0.719    | 0.077 | 0.066  | 0.947    | 0.122 | 0.065  | 0.538    | 0.071 | 0.139  |
| <i>internet</i>                  | 0.295    | 0.072 | 0.033  | 0.317    | 0.098 | 0.026  | 0.344    | 0.070 | 0.110  |
| <i>computer</i>                  | 0.769    | 0.097 | 0.095  | 1.035    | 0.145 | 0.095  | 0.320    | 0.064 | 0.101  |
| <i>formalcred</i>                |          |       |        |          |       |        | 0.140    | 0.037 | 0.042  |
| <i>gvacapital<sup>b</sup></i>    |          |       |        | -0.025   | 0.008 | -0.002 | -0.022   | 0.005 | -0.006 |
| <i>outputcapital<sup>f</sup></i> |          |       |        | -0.004   | 0.001 | 0.000  |          |       |        |
| <i>gvaturnover<sup>d</sup></i>   | -2.420   | 0.251 | -0.254 |          |       |        |          |       |        |
| <i>Jammu &amp; Kashmir</i>       | -1.282   | 0.156 | -0.095 | -1.410   | 0.209 | -0.083 | -0.716   | 0.118 | -0.166 |
| <i>Himachal Pradesh</i>          | 0.494    | 0.120 | 0.059  | 0.878    | 0.183 | 0.081  | 0.165    | 0.101 | 0.052  |
| <i>Punjab</i>                    | -1.009   | 0.131 | -0.080 | -1.147   | 0.179 | -0.071 | -0.329   | 0.102 | -0.089 |
| <i>Uttarakhand</i>               | -0.500   | 0.123 | -0.045 | -0.470   | 0.166 | -0.033 | -0.069   | 0.134 | -0.020 |
| <i>Haryana</i>                   | -0.575   | 0.106 | -0.051 | -0.694   | 0.148 | -0.047 | -0.054   | 0.099 | -0.016 |
| <i>Delhi</i>                     | -0.528   | 0.104 | -0.048 | -0.902   | 0.164 | -0.058 | -0.200   | *     | -0.056 |

|                                                  |        |       |        |             |     |        |        |            |       |        |
|--------------------------------------------------|--------|-------|--------|-------------|-----|--------|--------|------------|-------|--------|
| Rajasthan                                        | ***    | 0.111 | -0.057 | -0.870      | *** | 0.163  | -0.057 | -0.062     | 0.106 | -0.018 |
| Uttar Pradesh                                    | **     | 0.084 | -0.019 | -0.058      |     | 0.115  | -0.004 | 0.071      | 0.086 | 0.021  |
| Bihar                                            | ***    | 0.137 | -0.081 | -1.565      | *** | 0.226  | -0.088 | -0.460     | ***   | 0.143  |
| Assam                                            | ***    | 0.172 | -0.099 | -2.123      | *** | 0.286  | -0.103 | -0.781     | ***   | 0.165  |
| West Bengal                                      | ***    | 0.176 | -0.114 | -2.274      | *** | 0.280  | -0.115 | -0.840     | ***   | 0.087  |
| Orissa                                           | ***    | 0.123 | -0.046 | -0.864      | *** | 0.189  | -0.056 | -0.294     | ***   | 0.112  |
| Chhattisgarh                                     | ***    | 0.190 | -0.104 | -1.753      | *** | 0.254  | -0.093 | -0.775     | ***   | 0.140  |
| Madhya Pradesh                                   | ***    | 0.133 | -0.084 | -1.129      | *** | 0.177  | -0.070 | -0.123     | ***   | 0.129  |
| Gujarat                                          | ***    | 0.150 | -0.092 | -1.358      | *** | 0.199  | -0.080 | -0.667     | ***   | 0.102  |
| Maharashtra                                      | ***    | 0.165 | -0.106 | -1.907      | *** | 0.247  | -0.101 | -0.573     | ***   | 0.084  |
| Karnataka                                        | ***    | 0.106 | -0.050 | -0.630      | *** | 0.147  | -0.043 | -0.261     | ***   | 0.083  |
| Kerala                                           | ***    | 0.095 | -0.013 | -0.300      | **  | 0.136  | -0.022 | -0.060     | ***   | 0.081  |
| Tamil Nadu                                       | ***    | 0.113 | -0.065 | -1.053      | *** | 0.175  | -0.066 | -0.445     | ***   | 0.079  |
| _cons                                            | ***    | 0.500 |        | -17.960     | *** | 1.979  |        | -1.714     | ***   | 0.103  |
| Insignia2                                        |        |       |        |             |     |        |        |            |       |        |
| Inmktasset                                       | ***    | 0.007 |        |             |     |        |        |            | ***   | 0.025  |
| Intotwork                                        |        |       |        | 0.030       | **  | 0.014  |        | 0.060      |       |        |
| Inturn                                           |        |       |        | 0.065       | *** | 0.008  |        |            |       |        |
| No. of Observations                              |        |       |        | 58,765      |     |        |        | 10,606     |       |        |
| Wald chi <sup>2</sup>                            | df: 30 |       |        | 91.370      | *** | df: 33 |        | 1,053.100  | ***   | df: 31 |
| Log likelihood                                   |        |       |        | -19,504.650 |     |        |        | -4,512.204 |       |        |
| Pseudo R <sup>2</sup>                            |        |       |        | 0.124       | *** |        |        | 0.256      |       |        |
| LR chi <sup>2</sup> for Ho:Insignia <sup>2</sup> | df: 1  |       |        | 120.050     |     | df: 2  |        | 6.100      | **    | df: 1  |

**Source:** Computed by authors based on NSSO 73rd round unit-level data.

**Note:** \*\*\*, \*\*, \* imply the significance level of the estimated z statistic at the 0.01, 0.05 and 0.10 levels, respectively. SE is standard error; ME is marginal effect; df is degree of freedom.

<sup>a</sup> Dummy variable 1 if the enterprise has taken any assistance from the government which includes subsidies, financial loans, machinery and equipment, marketing, training, raw materials and others and 0 otherwise.

<sup>b</sup> Ratio of gross value added to market value of total assets

<sup>c</sup> Ratio of annual turnover to market value of total assets

<sup>d</sup> Ratio of annual gross value added to annual turnover/receipts

## Notes

1. [http://www.gstcouncil.gov.in/sites/default/files/newsletter/GSTC%20Secretariat%20News%20Letter\\_July%202020.pdf](http://www.gstcouncil.gov.in/sites/default/files/newsletter/GSTC%20Secretariat%20News%20Letter_July%202020.pdf).
2. For more details, please refer to the Key Indicator Report of NSS 73rd Round Survey (NSSO, 2017).
3. We find that only 1.4% of the sample enterprises (after applying all data restrictions) with an annual turnover below the VAT registration threshold are registered (Table A3).
4. Own-account enterprises (OAEs) are those which are run without any hired worker employed on a fairly regular basis. An establishment is one which employs at least one hired worker on a fairly regular basis. Within establishments, non-directory establishments (NDEs) have one to five workers (household and hired taken together) and directory establishments (DEs) have six or more workers (household and hired taken together).
5. Having access to internet or computer means the use of internet or computer by the enterprises during the last 365 days.
6. Total assets include market value of owned and hired assets excluding land and including building, plant and machinery, transport equipment, tools and other fixed assets, software and database, information and communications technology (ICT) equipment and capital work in progress. The value of total assets determines the capital base of the enterprises.
7. We have used Stata (version 14.0) command 'hetprobit' for estimating the probit model.
8. Formal sources of loan includes central and state-level term lending institutions, government (central, state and local bodies), commercial banks, co-operative banks and societies, micro-finance institutions and other institutional agencies. Informal sources include loans from the money lenders, business partner(s), suppliers/contractors, friends and relatives and other sources of loan.

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